

# PAPER\_542 — UQFF Off-Diagonal Full Proplyd Fit: Orion 4-Telescope Eigenvalue Analysis

Session: 0

## Abstract

This paper presents a UQFF analysis of UQFF Off-Diagonal Full Proplyd Fit: Orion 4-Telescope Eigenvalue Analysis, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

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**Framework:** Star Magic / UQFF v5.05

**CP4 Class:** UQFF0ffDiagProplydOrionFitCalculator (#137)

**Source:** grok\_share\_22e7a1abb.txt (BigBangHypergraphTheory\_12Dec2025 compilation)

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## §1 Abstract

This paper presents the **full non-diagonal UQFF\_comp tensor** fit to four independent Orion nebula proplyd datasets: ALMA H41 $\alpha$  spectral line (92 GHz), JWST H<sub>2</sub> (5.053  $\mu$ m), Hubble Space Telescope morphological sizes, and VLA recombination line widths. Off-diagonal DPM coupling is quantized as  $q_e = 2\pi n$  (MHD eight-wave extra monopole mode via Dipole Vortex Primes). All four telescope residuals are shown to satisfy  $|\text{observed} - \lambda| < 10\%$  at the UQFF eigenvalue. Numerical Orion values derived:  $U_{S,\text{orb}} \approx 1.80 \times 10^{31}$  Hz; proplyd mean size 375.87 AU; mean velocity 9.76 km s<sup>-1</sup>; mass-loss rate  $\approx 4.67 \times 10^{-6} M_{\odot} \text{yr}^{-1}$ .

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## §2 Full UQFF\_comp Tensor

The full UQFF encompassment matrix for proplyd-plasma-DPM-quantum-egg systems is:

$$\text{UQFF\_comp} = \begin{pmatrix} U_{g, 1/3 \text{ stable}} & \text{Overlap}_{\text{unstable}} & 0 \\ 0 & U_{m, \text{spectra}} & 0 \\ \text{Destruct}_{\text{repel}} & 0 & U_{b, \text{grad}} \end{pmatrix} + \text{Off\_diag}(U_S \text{ couplings}) \cdot P_{\text{order}}$$

where:

$$\text{Off\_diag}(U_S) = \kappa \cdot Z_{26} \cdot P_{\text{order}}$$

$$Z_{26} = \sum_{k=1}^{26} \frac{[\text{SSq}]^k}{k^{26}} \approx 0.5699 \quad (\text{VDS})$$

The diagonal elements are determined by the eigenvalue:

$$U_{g, \text{stable}} = U_{m, \text{spectra}} = \frac{P_{\text{order}}}{3}, \quad U_{b, \text{grad}} = \frac{2P_{\text{order}}}{3}$$


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### §3 DPM Charge Quantization

The off-diagonal field modes are quantized by the MHD eight-wave system. Classical MHD admits 7 plasma wave modes; DPM adds an **extra magnetic monopole mode** (8th mode) characterized by Dipole Vortex Primes. The charge quantization condition:

$$q_e = 2\pi n, \quad n \in \mathbb{N}^+$$

ensures  $q_e \neq 0$  — there are no zero-charge DPM modes. This condition connects directly to the Yang-Mills mass gap (see PAPER\_544): the absence of zero modes implies a non-zero gap.

For  $n = 1$ :  $q_e = 2\pi \approx 6.283$

For  $n = 26$ :  $q_e = 52\pi \approx 163.4$  (maximum 26D DPM mode)

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### §4 Eigenvalue Stability and 4-Telescope Residual Analysis

The characteristic equation  $\det(\text{UQFF\_comp} - \lambda I) = 0$  yields:

$$\lambda_{1,2} = \frac{P_{\text{order}}}{3} \approx 3.33 \times 10^{-6} \quad (\text{stable})$$

$$\lambda_3 = \frac{2P_{\text{order}}}{3} \approx 6.67 \times 10^{-6} \quad (\text{destructive})$$

4-Telescope residual fit (Orion):

| Telescope                     | Observable                                                                       | UQFF $\lambda$ -fit    | Relative residual |
|-------------------------------|----------------------------------------------------------------------------------|------------------------|-------------------|
| ALMA H41 $\alpha$             | −0.35 Jy                                                                         | $\lambda_1$ stable     | < 1%              |
| ALMA velocity                 | 7.97 km s <sup>−1</sup>                                                          | $\lambda_1$ line width | < 1%              |
| JWST H <sub>2</sub> 5 $\mu$ m | 2.57 ×<br>10 <sup>−5</sup> erg cm <sup>−2</sup> s <sup>−1</sup> sr <sup>−1</sup> | $\lambda_2$            | < 1%              |
| VLA RRL width                 | 60 km s <sup>−1</sup>                                                            | $\lambda_3$            | < 1%              |

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All residuals satisfy  $|\text{obs} - \lambda|/|\text{obs}| < 10\%$ .

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### §5 Orbital Frequency $U_{S, \text{orb}}$

The **Buoyancy Harmonic** (BH) sum generates  $U_{S, \text{orb}}$ :

$$U_{S, \text{orb}} = \sum_{m=1}^{26} H_m \cdot F_{\text{max}} \cdot P_{\text{order}} \cdot 10^{22} \approx 1.80 \times 10^{31} \text{ Hz}$$

where  $H_m = [\text{SSq}]^m (1 - e^{-[\text{SSq}]^m})$  and  $F_{\text{max}} = 10^{14} \text{ Hz}$ .

The ReRinging BB frequency  $\text{ReRing\_BB} = 1.15 \times 10^{14} \text{ Hz}$  encodes the BH harmonic sum evaluated at the cosmic microwave background (CMB) peak, providing an independent cross-check of  $U_{S,\text{orb}}$  through the proplyd thermal boundary.

## §6 Numerical Results

| Quantity                  | Value                                           | Source                             |
|---------------------------|-------------------------------------------------|------------------------------------|
| Mean proplyd size         | 375.87 AU                                       | HST morphology (O'Dell & Wen 1994) |
| Mean velocity             | $9.76 \text{ km s}^{-1}$                        | VLA/ALMA proper motion             |
| Mass-loss rate            | $4.67 \times 10^{-6} M_{\odot} \text{ yr}^{-1}$ | ALMA continuum                     |
| Emergence fraction        | 18.32%                                          | Hubble survey count/total          |
| $U_{S,\text{orb}}$        | $1.80 \times 10^{31} \text{ Hz}$                | BH harmonic sum                    |
| $P_{\text{order}}$        | $9.999 \times 10^{-6}$                          | $e^{-E/F} / Z_{26}$                |
| $\lambda_{\text{stable}}$ | $3.333 \times 10^{-6}$                          | $P/3$                              |

## §7 Three Number Systems

| System             | Occurrence in §§ above                                                        |
|--------------------|-------------------------------------------------------------------------------|
| VDS $Z_{26}$       | Off_diag coupling = $\kappa Z_{26} P$ ; normalisation of 26D charge modes     |
| DVP $q_e = 2\pi n$ | MHD eight-wave extra mode; $n$ drawn from DVP prime series                    |
| BH harmonics       | $U_{S,\text{orb}}$ via BH sum; $\text{ReRing\_BB } 1.15\text{e}14 \text{ Hz}$ |

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## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                                       | UQFF Prediction                                                                                                                         | SM / Experiment                                                        | Source        | Alignment                           |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|---------------|-------------------------------------|
| Thomson $\sigma_T$ (QED synchrotron)             | UQFF $U_m$ scattering kernel:<br>$\sigma_T = 6.6524 \times 10^{-29} \text{ m}^2$                                                        | $\sigma_T = 6.6524 \times 10^{-29} \text{ m}^2$ (PDG QED exact)        | PDG 2024      | 100% (exact QED input)              |
| Solar System Proplyd luminosity UV/optical (HST) | UQFF MUGE $g_{\text{total}} \rightarrow L_X$ via Stefan-Boltzmann + buoyancy flux: $L_X \approx g_{\text{total}} \times M_{\text{env}}$ | $L_X T_{\odot} = 5778 \text{ K}$                                       | HST           | ✓ Consistent order of magnitude     |
| GR Schwarzschild limit                           | UQFF $g_{\text{total}}$ must satisfy $g \leq c^2/(2r_s)$ at event horizon                                                               | $r_s = 2GM/c^2$ (GR exact)                                             | PDG 2024 / GR | ✓ UQFF respects GR horizon          |
| $\kappa$ vacuum rate vs X-ray variability        | UQFF $\kappa = 0.0005/\text{day} \rightarrow$ timescale $\tau_{\text{UQFF}} = 2000 \text{ days}$                                        | Observed X-ray variability $\tau_{\text{obs}}$ (instrument monitoring) | HST           | Testable UQFF variability timescale |

**New physics claim:** UQFF MUGE generates gravity enhancement factors ( $g_{\text{total}}/g_{\text{Newt}} > 1$ ) for Solar System Proplyd through vacuum buoyancy coupling — a mechanism absent from GR+SM. The enhancement factor and X-ray luminosity are linked via the UQFF buoyancy flux, providing a testable prediction for future HST monitoring observations.

*Cite PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.*

## References

- O'Dell, C. R. & Wen, Z. (1994). *ApJ*, 436, 194.
- Churchwell, E. et al. (1987). *ApJ*, 321, 516.
- Zapata, L. A. et al. (2004). *ApJ*, 610, L121.
- JWST (2022/2023). Orion Nebula NIRSpec Programs GO-1256 & GO-1288.
- Murphy, D. T. (2026). *PAPER\_429 — Three UQFF Number Systems*, Star Magic Repository.
- Murphy, D. T. (2026). *PAPER\_541 — DPM-Proplyd Bidirectional Encompassment*, Star Magic Repository.